Supplementary material

Supplementary Table 1 Search strategy using PubMed.

Search number	Query	Results
1	"Cholangiopancreatography, Endoscopic Retrograde"[Mesh] OR endoscopic-retrograde-cholangiopancreatography OR endoscopic-cholangiography OR endoscopic-cholangiopancreatography OR endoscopic-pancreatocholangiography OR endoscopic-pancreatography OR endoscopic-retrograde-cholangiography OR endoscopic-retrograde-pancreaticocholangiography OR "ERCP" OR retrograde-cholangiopancreatography OR retrograde-endoscopic-cholangiography OR retrograde-endoscopic-cholangiopancreatography”	24730
2	guidewire* OR guide-wire* OR wire-guide*	10006
3	(Randomized-controlled-trial[pt] OR controlled-clinical-trial[pt] OR randomized[tiab] OR placebo[tiab] OR clinical-trials-as-topic[mesh: noexp] OR randomly[tiab] OR trial[ti])	1499874
4	#1 AND #2	658
5	#3 AND #4	97
6	#5 NOT ("animals"[mesh] NOT "humans"[mesh])	97
7	#6 NOT ("case reports"[Publication Type] OR "comment"[Publication Type] OR "editorial"[Publication Type] OR "guideline"[Publication Type] OR "introductory journal article"[Publication Type] OR "meta analysis"[Publication Type] OR "news"[Publication Type] OR "retracted publication"[Publication Type] OR "review"[Publication Type] OR "systematic review"[Publication Type])	79

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Supplementary material

Supplementary Table 2 Risk of bias using Cochrane tool for randomized controlled trials.

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)
Bassan et al., 2018	Low	Low	High	Low	Low	Low
Halttunen et al., 2012	Unclear	Unclear	High	Low	Low	Low
Kitamura et al., 2015	Low	Unclear	High	Low	Low	Low

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Supplementary material

Supplementary Table 3 EMBASE search strategy.

Search number	Query	Results
1	'endoscopic retrograde cholangiopancreatography'/exp OR 'endoscopic retrograde cholangiopancreatography cannula'/exp OR 'endoscopic retrograde cholangiopancreatography' OR 'endoscopic cholangiography' OR 'endoscopic cholangiopancreatography' OR 'endoscopic pancreatocholangiography' OR 'endoscopic pancreatography' OR 'endoscopic retrograde cholangiography' OR 'endoscopic retrograde pancreaticocholangiography' OR 'ercp' OR 'retrograde cholangiopancreatography' OR 'retrograde endoscopic cholangiography' OR 'retrograde endoscopic cholangiopancreatography'	47961
2	guidewire* OR 'guide wire*' OR 'wire guide*' OR 'guide wire'/exp	35447
3	random* OR factorial* OR crossover* OR 'cross over' OR placebo* OR (doubl* NEXT/2 blind*) OR (singl* NEXT/2 blind*) OR assign* OR allocat* OR volunteer* OR 'crossover procedure'/exp OR 'double-blind procedure'/exp OR 'randomized controlled trial'/exp OR 'single-blind procedure'/exp	2854351
4	#1 AND #2	2695
5	#3 AND #4	310
6	#5 NOT ([animals]/lim NOT [humans]/lim)	308
7	#6 NOT ('conference review'/it OR 'editorial'/it OR 'letter'/it OR 'note'/it OR 'review'/it OR 'short survey'/it OR 'tombstone'/it OR 'case report'/de OR 'meta analysis'/de OR 'meta analysis topic'/de OR 'practice guideline'/de OR 'systematic review'/de OR 'systematic review topic'/de)	220

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Supplementary material

Supplementary Table 4 Cochrane Central search strategy.

Search number	Query	Results
1	[mh "Cholangiopancreatography, Endoscopic Retrograde"] OR endoscopic-retrograde-cholangiopancreatography OR endoscopic-cholangiography OR endoscopic-cholangiopancreatography OR endoscopic-pancreatocholangiography OR endoscopic-pancreatography OR endoscopic-retrograde-cholangiography OR endoscopic-retrograde-pancreaticocholangiography OR "ERCP" OR retrograde-cholangiopancreatography OR retrograde-endoscopic-cholangiography OR retrograde-endoscopic-cholangiopancreatography	2490
2	guidewire* OR guide-wire* OR wire-guide*	1355
3	#1 AND #2	211
4	randomized OR placebo OR [mh ^"clinical trials as topic"] OR randomly OR trial	1532961
5	#3 AND #4	202

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Supplementary material

Supplementary Table 5 Web of Science search strategy.

Search number	Query	Results
1	random* OR factorial* OR crossover* OR cross-over OR placebo* OR double-blind* OR single-blind* OR doubly-blind* OR singly-blind* OR assign* OR allocat* OR volunteer* (Title) or random* OR factorial* OR crossover* OR cross-over OR placebo* OR double-blind* OR single-blind* OR doubly-blind* OR singly-blind* OR assign* OR allocat* OR volunteer* (Abstract) or random* OR factorial* OR crossover* OR cross-over OR placebo* OR double-blind* OR single-blind* OR doubly-blind* OR singly-blind* OR assign* OR allocat* OR volunteer* (Author Keywords)	3028583
2	guidewire* OR guide-wire* OR wire-guide* (Title) or guidewire* OR guide-wire* OR wire-guide* (Abstract) or guidewire* OR guide-wire* OR wire-guide* (Author Keywords)	9211
3	endoscopic-retrograde-cholangiopancreatography OR endoscopic-cholangiography OR endoscopic-cholangiopancreatography OR endoscopic-pancreatocholangiography OR endoscopic-pancreatography OR endoscopic-retrograde-cholangiography OR endoscopic-retrograde-pancreaticocholangiography OR "ERCP" OR retrograde-cholangiopancreatography OR retrograde-endoscopic-cholangiography OR retrograde-endoscopic-cholangiopancreatography (Title) or endoscopic-retrograde-cholangiopancreatography OR endoscopic-cholangiography OR endoscopic-cholangiopancreatography OR endoscopic-pancreatocholangiography OR endoscopic-pancreatography OR endoscopic-retrograde-cholangiography OR endoscopic-retrograde-pancreaticocholangiography OR "ERCP" OR retrograde-cholangiopancreatography OR retrograde-endoscopic-cholangiography OR retrograde-endoscopic-cholangiopancreatography (Abstract) or endoscopic-retrograde-cholangiopancreatography OR endoscopic-cholangiography OR endoscopic-cholangiopancreatography OR endoscopic-pancreatocholangiography OR endoscopic-pancreatography OR endoscopic-retrograde-cholangiography OR endoscopic-retrograde-pancreaticocholangiography OR "ERCP" OR retrograde-cholangiopancreatography OR retrograde-endoscopic-cholangiography OR retrograde-endoscopic-cholangiopancreatography (Author Keywords)	17279
4	#2 AND #3	557
5	#1 AND #4	103
6	#1 AND #4 and Review Articles or Editorial Materials (Exclude – Document Types)	95

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Supplementary material

Supplementary Table 6 Global Index Medicus search strategy.

Search number	Query	Results
1	tw:((tw:("endoscopic retrograde cholangiopancreatography" OR "endoscopic cholangiography" OR "endoscopic cholangiopancreatography" OR "endoscopic pancreatocholangiography" OR "endoscopic pancreatography" OR "endoscopic retrograde cholangiography" OR "endoscopic retrograde pancreaticocholangiography" OR "ERCP" OR "retrograde cholangiopancreatography" OR "retrograde endoscopic cholangiography" OR "retrograde endoscopic cholangiopancreatography")) AND (tw:(guidewire* OR "guide wire" OR "wire guided")) AND (tw:(random* OR factorial* OR crossover* OR "cross over" OR placebo* OR "double blind" OR "single blind" OR "doubly blind" OR "singly blind" OR assign* OR allocat* OR volunteer*)))	14

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Supplementary material

Section and Topic	Item #	Checklist item	Location where item is reported (page no.)
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	7, 8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary material
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7, 8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7, 8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7, 8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	8, 9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	7, 8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8, 9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	8, 9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	8, 9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	8, 9

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Supplementary material

Section and Topic	Item #	Checklist item	Location where item is reported (page no.)
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	10, Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	10, references
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	11, Supplementary table 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	10, 11
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	10, 11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	10, 11
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	10, 11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	11
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	11
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	12
	23b	Discuss any limitations of the evidence included in the review.	13
	23c	Discuss any limitations of the review processes used.	13
	23d	Discuss implications of the results for practice, policy, and future research.	13
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	NA
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	2

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Supplementary material

Section and Topic	Item #	Checklist item	Location where item is reported (page no.)
Competing interests	26	Declare any competing interests of review authors.	2
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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